Advancing the INFORM Hazard Evidence for Tanzania while Preserving the INFORM Engine: a CHIRPS/ERA5/NBS multi-criteria drought case study

INFORM Tanzania (inform.co.tz)

2026-06-03, corrected 2026-10-09

> **Status, corrected 9 October 2026.** This paper (June 2026) proposed advancing the hazard evidence under a “raise-only” rule and quoted national figures from the working dataset. The integrity assessment of October 2026 (`docs/assessment/INFORM_TZ_Integrity_Assessment_2026-10.md`) superseded both, and the corrections are recorded here so that no copy of this paper circulates without them:
>
> 1. **The national figures in Section 2** (Hazard 2.2, Vulnerability 5.5, Lack-of-Coping 5.9, Risk 4.1) were the **Kondoa district row** of the workbook, not a national value. The INFORM aggregate of the 195 councils on the workbook data is **3.8** (Hazard & Exposure 2.4, Vulnerability 3.8, Lack of Coping Capacity 6.2), and INFORM publishes no official national figure for a subnational model.
> 2. **Raise-only is withdrawn.** The Core carries the workbook values exactly (built by `scripts/build-core.mjs`, reproduced at 8,664/8,664 standardised values and 170/170 risks). Earth observation evidence enters a separate **Advancement layer** as weighted sub-indicators beside the workbook member of each indicator group, over the members with data, and is shown beside the Core on every profile, never in its place. Nothing floors, raises or lowers a Core value.
> 3. **The blended drought composite of Section 3 is withdrawn.** It scored wet highland councils (Lushoto) 10 by selecting the erratic short rains as the main season. Aridity, SPEI depth, rainfall variability, NDVI and soil moisture now enter the Advancement layer as separate members with their sources; the growing-season member is withheld until both rainy seasons are scored (issue MET-01).
> 4. **The flood = max(H, sqrt(H x E)) rule of Section 6 is removed** from the model. Heavy-rainfall frequency enters the Advancement layer; exposure is shown as the 2022 census population, not multiplied into any hazard.
> 5. **Earthquake** in the Core is the workbook’s GEM exposure index; the USGS catalogue (events within 50 km) is an Advancement member, not a Core source.
>
> The Tier 1 engine-fidelity items of Section 8 are closed by the Core rebuild. The body below is kept as the record of the June 2026 proposal; where it states a method no longer used, the status is marked in brackets.

> **Original status note (June 2026) - proposal for INFORM Calc Engine v2.** The operational model (**v1**) keeps the INFORM aggregation engine **exactly** as the official Excel workbook - verified formula-by-formula, written in the code in the workbook’s own literal form (product `GEOMEAN`, product of cube roots, `AVERAGE`, min-max). The

> advancements documented in this paper are a **proposal**; once validated ("when it passes") they are implemented as **INFORM Calc Engine v2**. In short: **v1 = exact INFORM now; v2 = these advancements next**. This is one of three documents - (1) the operational **Manual**, (2) the **Exact-formula reference** (`INFORM_CALCULATION_ENGINE.docx`, what v1 runs), and (3) this **Paper** (the v2 proposal).

Abstract

The INFORM Subnational Risk Index aggregates standardised (0-10) hazard, vulnerability and lack-of-coping indicators into a comparable risk score. Its aggregation **engine** - min-max standardisation (with logarithmic transformation and Tukey outlier capping), arithmetic averaging of indicators into categories, the **scaled geometric mean** of categories into dimensions, and the **cube root** of the three dimensions into risk - is a fixed methodology we **preserve exactly** as implemented in the official *Tanzania - Country Model Template* workbook (verified formula-by-formula from its hidden sheets). What we **advance** is the *evidence* behind individual hazard indicators where authoritative national data exist. This paper documents the advancement of the **drought hazard** from a coarse INFORM SADC value to a transparent four-component index computed from CHIRPS v3 rainfall (1991-2024) and ERA5 temperature - aridity, interannual variability, SPEI drought depth, and growing-season failure - together with the **initial vs advanced** comparison, the **impact**, and the **gap** the advancement closes. We show the initial drought layer was spatially incoherent (the central semi-arid zone under-rated at ~4/10, a coastal city over-rated at 10/10), whereas the advanced layer tracks observed rainfall and the food-security geography. The flood, heatwave and exposure advancements are summarised. Finally we open a **living, four-tier v2 issue register** (Section 8) - engine-fidelity, hazard-evidence, exposure/vulnerability resolution, and validation - and the three cross-cutting insights (stable references, hazard-specific exposure, outcome validation) that anchor it (Section 9). Every item is a proposal for a future, separately-validated INFORM Calc Engine v2; none changes the v1 engine running today.

1. Introduction

INFORM (the Index for Risk Management; DRMKC/JRC, European Commission) provides a transparent, reproducible composite risk model [1]. For Tanzania, the operational instrument is the *Tanzania - Country Model Template* workbook, whose engine standardises each indicator to 0-10 and aggregates them. The engine is deliberately conservative and globally comparable; it must not be silently altered.

Our work separates two concerns:

1. **The engine** (how numbers combine). This stays **identical** to the workbook - we verified every formula directly from the (hidden) engine sheets (see the companion `INFORM_CALCULATION_ENGINE.docx`): component = `AVERAGE`, category = `AVERAGE`, dimension = `ROUND((10-GEOMEAN((10-c)/10*9+1...))/9*10,1)` (scaled geometric mean), risk = `ROUND(H^(1/3)*V^(1/3)*LCC^(1/3),1)` (cube root), and standardisation = min-max to 0-10

with optional log/exp transform, Tukey (Q1-1.5*IQR, Q3+1.5*IQR) outlier capping, and an inversion for “Decrease Risk” indicators.
2. **The evidence** (the value of each indicator). Where the INFORM SADC layer is coarse, dated, or incoherent for Tanzania, and authoritative national data exist, we **recompute the indicator** from primary data using the **same standardisation method** the engine prescribes, and we apply it **raise-only** (the computed value may raise a documented hazard but never silently lower it). *[Withdrawn October 2026: evidence enters the separate Advancement layer; the Core is never raised or lowered.]*

This paper makes the second concern explicit and auditable, using **drought** as the worked case, because drought is the hazard most consequential for Tanzanian food security and the one for which the densest observational record exists.

2. The preserved INFORM engine

We reproduced a unit end-to-end against the workbook: for Kondoa, the workbook’s own Hazard dimension is `scaled-geomean(Natural 3.544, Human 0.683) = 2.2`, and our implementation returns the same 2.2 for the same inputs; the figures we quoted as national (Hazard 2.2, Vulnerability 5.5, Lack-of-Coping 5.9, Risk 4.1) *[are Kondoa’s row, not a national value; corrected October 2026: the INFORM aggregate of the 195 councils is 3.8 (2.4, 3.8, 6.2)]*. The engine is therefore **unchanged**; only indicator *inputs* differed where we advanced them. Two engine-level options the workbook supports but our recompute scripts do not yet apply - **Tukey IQR capping** of recomputed series, and **custom (fixed) reference ranges** instead of the Tanzania data range - are recorded as **future work** (Section 7) and are not changed here.

3. The drought advancement - method

3.1 Rationale and the gap in the initial layer

The initial INFORM SADC drought scores for Tanzania are spatially incoherent (Section 5, Table 2): several central **semi-arid** councils (Kondoa 771 mm yr^-1, Singida 749 mm, Manyoni 760 mm) scored only 3.6-4.3/10, while a wet coastal city (Dar es Salaam / Ilala, 1,253 mm) scored 10/10. Semi-arid central Tanzania is a recurrent drought- and food-insecurity hotspot (IPC/MUCHALI classifications), so an index that rates it below 5 while maxing out a coastal city is not credible. This is the **gap**: the *initial* relevance of the drought layer was, in places, simply not true.

3.2 Data

- **CHIRPS v3** monthly precipitation, 0.05 deg, 1991-2024, the same product validated for Tanzanian rainfall classification [2]. Zonal-mean rainfall is extracted per administrative unit.
- **CHIRPS v3 daily** (2015-2024) for the companion heavy-rain/flood layer (Section 6).
- **ERA5** 2 m temperature, used to estimate potential evapotranspiration (PET) for SPEI [3].

3.3 The four-component index and its criteria

Drought relevant to communities is more than a single statistic; following the drought-index literature [4,5] we combine four physically distinct, individually justified components, each min-max scaled to 0-1:

Component	Weight	What it captures	Computation	Basis
Aridity	0.30	inherent dryness of the climate	min-max of **(-mean annual rainfall)**	UNEP aridity gradient; semi-arid = ~0.2-0.5 P/PET [6,7]
Interannual variability	0.20	year-to-year rainfall instability	min-max of the **coefficient of variation** of annual rainfall	East-African rainfall is among the world's most variable [8,9]
SPEI drought depth	0.15	severity of water-balance deficits	mean magnitude of **SPEI-12 < -1**, PET via Thornthwaite [10] from ERA5	SPEI = standardised precipitation-evapotranspiration index [3,11]
Growing-season failure	0.35	crop-failure / food-security driver	0.5*CV* + *0.5*frequency of the main 4-month season below 80% of normal	season failure -> IPC food crises (highest weight)

The blended score `0.30*aridity + 0.20*variability + 0.15*SPEI-depth + 0.35*season-failure` is then **min-max standardised to 0-10** - exactly the INFORM normalisation - and applied **raise-only** (the advanced value replaces the initial only where it is higher, so no documented drought area is downgraded). *[Withdrawn October 2026: the blend and the floor are out of the model; the members enter the Advancement layer separately and the season-failure member is withheld.]*

Weighting rationale. Aridity is the baseline (a dry climate is intrinsically drought-exposed); variability and SPEI add the *dynamic* deficit; **growing-season failure carries the largest weight (0.35)** because, for an operational risk index feeding humanitarian planning, the decision-relevant outcome is failure of the main cropping season, which is what triggers IPC/MUCHALI food-insecurity phases. The weights are an explicit modelling choice; a sensitivity analysis is listed as future work.

Why SPI/SPEI *frequency* is not used to rank. Drought *frequency* (the share of months below a threshold) is approximately stationary by construction across a fixed record and therefore discriminates poorly between districts; we use SPEI **depth** (how deep deficits go) rather than frequency, and rank on aridity + variability + season-failure, which are the spatially-informative signals.

4. Standardisation and integration (unchanged engine)

The four raw components are combined and the result is min-max scaled to 0-10 using the **same** formula the workbook applies to every indicator (`10*(x-Min)/(Max-Min)`, clamped,

rounded to 0.1). The drought score then enters the **unchanged** chain: it averages with the other natural-hazard components into the *Natural* category, which scaled-geometric-means with *Human* hazards into the *Hazard & Exposure* dimension, which cube-roots with Vulnerability and Lack-of-Coping into Risk. No engine constant is altered.

5. Results - initial vs advanced

Table 2. Initial (INFORM SADC) vs advanced (CHIRPS/ERA5) drought score, with the raw evidence.

District	Rainfall (mm yr^-1)	Rain CV	Season-fail freq	SPEI depth	Initial	Advanced
Longido (pastoral)	668	0.26	0.27	1.60	8.6	**9.7**
Monduli (pastoral)	694	0.26	0.27	1.56	8.6	**9.2**
Singida (semi-arid)	749	0.24	0.24	1.68	**3.6**	**8.7**
Manyoni (semi-arid)	760	0.20	0.18	1.63	**3.6**	**7.0**
Kondoa (semi-arid)	771	0.24	0.24	1.64	**4.3**	**8.6**
Rufiji (sub-humid)	1,088	0.21	0.21	1.50	2.9	5.8
Kilombero (wet)	1,598	0.17	0.21	1.45	0.7	2.9
Kyela (wettest)	1,996	0.16	0.21	1.48	0.0	2.2
Ilala / Dar (coastal city)	1,253	0.26	0.24	1.49	**10**	10*

*Dar es Salaam retains 10 under the raise-only floor; the computed value is ~6-7. This is a flagged artifact (the initial 10 is documented-but-wrong) queued for correction via Data Entry, *not* an advancement outcome. *[October 2026: the "initial" column of this table was the adjusted dataset, not the workbook; in the workbook Ilala's drought is the historic-frequency indicator like every other unit.]*

Pattern. The advanced layer is **monotone in rainfall**: the five driest councils (Longido, Monduli, Singida, Manyoni, Kondoa, all 668-771 mm) score 7.0-9.7; the wettest (Kyela, Kilombero, 1,600-2,000 mm) score 2.2-2.9. The initial layer was not monotone - it under-rated Singida/Manyoni/Kondoa (3.6-4.3) despite their identical semi-arid climate to the (correctly high) pastoral districts.

6. Other advancements (summary)

The same *initial -> advanced (same standardisation) -> raise-only* discipline was applied to:
[All withdrawn from the Core in October 2026; see Status.]

- **Flood / heavy rainfall.** Advanced from a static INFORM value to **CHIRPS v3 daily** counts of days > 50 mm and > 100 mm (2015-2024), combined with documented flood events and the INFORM **Hazard x Exposure** form `flood = max(hazard, sqrt(hazard * exposure))` so population exposure can only amplify a flood hazard, never hide it. Heavy-precipitation thresholds follow the ETCCDI extreme-rainfall indices.
- **Heatwave.** Advanced from blank/baseline to an **ERA5** temperature-climatology exposure proxy (mean + hottest-quarter), min-max capped - hot lowlands high, cool highlands low.
- **Exposure.** Advanced to **NBS 2022 PHC** population density (population / area), **log** min-max standardised - the workbook's own "Logarithm" transformation for the right-skewed density distribution.

Vulnerability (poverty, stunting, HIV+malaria, displacement, food insecurity) and parts of coping (WASH, health/education access, DRR) were similarly refreshed from NBS/TDHS/IPC/UNHCR sources. All used the engine's standardisation unchanged.

7. Discussion: initial vs advanced, impact, and the gap

Initial vs advanced. The advancement does not change *how* drought enters risk; it changes *what value* drought has, from a coarse/incoherent layer to one grounded in 34 years of observed rainfall and the water balance.

Impact / outcome. Risk rises where the evidence supports it (the central semi-arid belt and the pastoral north), in line with IPC/MUCHALI food-insecurity classifications, and the change is auditably raise-only *[no longer: the Core is the workbook; the Advancement layer is shown beside it]*. Because risk is a cube root of three dimensions, a drought rise of several points lifts the Hazard dimension modestly and the risk score by a fraction - appropriately, since drought is one of twelve natural hazards.

The gap. The advancement is justified precisely where **the initial relevance was not true**: Kondoa and Singida are textbook semi-arid, food-insecure districts that the initial layer scored below 5. The advancement closes that gap with transparent, reproducible evidence. Where the initial layer was already correct (pastoral Longido/Monduli), the advancement merely confirms and slightly raises it; where the initial layer was wrong-high (the Dar artifact), raise-only deliberately does **not** advance it down - that case is handled separately as a documented correction. *[October 2026: moot; the Core is the workbook and the Advancement layer is shown beside it.]*

8. The v2 roadmap - a structured issue register

The advancements above open a **living register** of issues to resolve in INFORM Calc Engine v2. Each is recorded with the same discipline: what the workbook (or INFORM) does, what v1 does today, the proposed v2 method, and the expected impact. The register runs in four tiers, from "make the standardisation literally match the workbook" to "advance the science of the hazard layers" to "prove the index predicts reality". Nothing here changes the v1 engine; each item is implemented only after separate validation.

Tier 1 - Engine fidelity (close the last standardisation gap with the workbook)

v1 reproduces the workbook's **aggregation** exactly - mean, scaled geomean, cube root, verified 0/195. The remaining gap is in the **standardisation** of our *recomputed* indicators (the companion `INFORM_CALCULATION_ENGINE.docx` flags A-G). v2 closes it:

#	Issue	Workbook	v1 today	v2 method	Impact
B	**Outlier capping**	Tukey Q1-1.5·IQR / Q3+1.5·IQR before min-max, a **per-indicator toggle**	no cap on recomputed series	replicate the workbook's exact outlier-detection set; cap to the fence	stops one extreme unit stretching a Tanzania-relative min-max
D/I	**Reference range and temporal stability**	Data Range **or Custom fixed** reference	Data Range over Tanzania	adopt the per-indicator Custom references; use **fixed** references for anything tracked over time	a unit's score stops drifting when *other* units update - the key to trend monitoring
A	**Denominators**	explicit per-capita / per-area denominator	rates built directly	verify each denominator matches the workbook	small, indicator-specific
C	**Transform set**	per-indicator Log / Exp / None	Log for density only	match Log/Exp/None indicator-by-indicator	corrects skew handling per layer
F	**Indicator use-set**	`useIndicator = Yes/No` filter	null-exclusion	reconcile the exact included-indicator list per dimension	which leaves count in a category
G	**Class breakpoints**	per-dimension threshold table	global risk bands	confirm the exact per-dimension breakpoints	class/colour only, not the score

Tier 1b - sparse-data fidelity and engine convergence (June 2026 audit)

A code audit of the two engines - the live `riskModel` and the formal `informCalculationEngine` - surfaced four fidelity points on sparse or degenerate data. Two were aligned to the workbook immediately (behaviour-neutral on the fully-populated dataset); two remain v2 decisions.

Finding	Workbook	Was	Now / v2
Dimension = 0 -> risk	0 (`0^(1/3)=0`)	formal engine returned null via a `>0` guard; the live engine already returned 0	**Resolved** - guard dropped, both engines now return 0

Finding	Workbook	Was	Now / v2
Category weighting	unweighted (equal)	formal engine passed 0.5/0.5 weights into the geomean (identical at equal, divergent if changed)	**Resolved** - dimension geomean hardwired unweighted; weighting is a flagged v2 deviation only
Missing whole category	category `AVERAGE` errors, dimension blanks	both engines degrade to the single present category	**v2 decision** - replicate the blank (strict) or keep the graceful fallback and document it
Two engines of record	one workbook	live `riskModel` and formal `informCalculationEngine` agree on populated data, diverge on edge cases	**Partly** - the workbook-golden fixture now guards **both**; converging to one engine is v2

Equal weights and full data masked all four; only an explicit audit against the workbook's own sparse-data behaviour exposes them. The golden fixture (the workbook's cached rows) is now the regression guard pinning **both** engines to the workbook.

Tier 2 - Hazard evidence (apply the drought template to every documented overlay)

Several hazards are still **documented overlays** (expert-placed, not computed). v2 advances each the way drought was advanced - computed from an authoritative observational product, standardised by the unchanged engine, applied **raise-only** *[October 2026: as members of the Advancement layer, never as floors on the Core]*:

Hazard	v1 overlay	v2 evidence	Standard metric	Source
Earthquake	rift-proximity	probabilistic seismic hazard	PGA at 475-yr return period	USGS / GEM / GSHAP [13]
Flood	event counts × exposure	return-period inundation	1-in-100-yr flood extent/depth	JRC GloFAS / global flood maps [14]
Wildfire	miombo dry-season	active-fire + burned-area climatology	annual fire density	MODIS/VIIRS MCD64 [15]
Conflict / violence	documented	event intensity and fatalities	ACLED event rate	ACLED [16]
Lightning	Lake Victoria overlay	flash-density grid	flashes km^-2 yr^-1	NASA LIS/OTD [17]
Storms / cyclone	documented coastal	track density / return period	cyclone passage frequency	IBTrACS [18]
Heatwave	ERA5 climatology, capped	heat-health threshold days	days > 90th-pct Tmax; WBGT	ERA5 + heat-health [19]

Tier 3 - Exposure and vulnerability resolution

Issue	v1 today	v2 method
Hazard-specific exposure	one population-density layer multiplies every hazard	pair each hazard with its own exposure - drought with cropland, livestock and rural population; flood with floodplain population - not generic density

Issue	v1 today	v2 method
Council-resolution vulnerability	195 councils inherit district survey values	small-area estimation (model-based) for council-level vulnerability where surveys allow
Live food security	IPC/MUCHALI entered per round	automated IPC-round ingestion so livelihoods refreshes itself

Tier 4 - Methodology and validation (scientific credibility)

Issue	v1 today	v2 method
Reliability gating	data-coverage % shown, not enforced	a confidence tier per unit; annotate or suppress scores below a minimum coverage
Uncertainty	point scores	propagate input uncertainty (CHIRPS error, survey sampling) to a risk **band**
Validation	face validity vs IPC geography	quantitative validation against observed losses and displacement (EM-DAT, DesInventar Tanzania) [20]
Sensitivity	drought weights an explicit choice	full sensitivity analysis of which indicators and weights move risk
Weighting (deviation-flagged)	equal weights within a category (INFORM default)	explore expert/data-driven weights as a **separate, validated** option - never silently

9. Cross-cutting insights for v2

Three insights run across the register and shape the v2 design.

1. Temporal comparability is a reference-range problem. A Data-Range min-max re-standardises *every* unit whenever *any* unit's data changes, so a district's score can move even when its own reality did not. For a model meant to answer "is this place getting riskier?", that is a defect. v2's adoption of **fixed reference ranges** (Tier 1, #D/I) is therefore not cosmetic - it is what makes year-on-year and pre/post-intervention comparison valid. This is the single most consequential engine-fidelity decision.

2. Exposure must be hazard-specific or it misleads. v1 multiplies hazards by one population-density layer. But drought's exposure is **agricultural** - cropland, livestock, rural livelihoods - not urban population; a dense coastal city is highly *flood* and *heat* exposed but not *drought* exposed. Using a single exposure layer is exactly what produced the Dar-es-Salaam drought artifact (Table 2). Hazard-specific exposure (Tier 3) is the structural fix behind that symptom, not a cosmetic one.

3. A risk index is only as credible as its validation. The drought advancement is justified by face validity against IPC/MUCHALI geography; that is necessary but not sufficient. v2 must validate the risk score **quantitatively** against independent outcomes - recorded disaster losses, displacement, food-insecurity caseloads - so "risk" is demonstrably predictive, not merely plausible. Without this, every advancement is an assertion; with it, the model earns operational trust.

Stable references, hazard-specific exposure, and outcome validation are the backbone of INFORM Calc Engine v2.

10. Limitations of this proposal

- **Weights** for the drought blend (0.30/0.20/0.15/0.35) are an explicit modelling choice; the sensitivity analysis (Tier 4) is what will defend or revise them.
- **Raise-only** is honest but asymmetric: it cannot correct a documented-but-too-high value (the Dar artifact), which is handled separately as a Data-Entry correction, not an advancement. *[Resolved by withdrawing the rule: the Core is the workbook and the Advancement layer is shown beside it.]*
- The **register is not exhaustive** and is expected to grow; it is a working backlog, not a finished spec.
- Every Tier-2/3/4 item is **unvalidated** until its own evidence passes review - listing it here is a commitment to test it, not a claim that it is already right.

References

[1] INFORM (2024). *INFORM Subnational Risk Index - SADC*. Disaster Risk Management Knowledge Centre / Joint Research Centre, European Commission. [2] Funk, C. et al. (2015). The climate hazards infrared precipitation with stations - a new environmental record for monitoring extremes. *Scientific Data* 2, 150066. [3] Hersbach, H. et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society* 146, 1999-2049. [4] Zargar, A. et al. (2011). A review of drought indices. *Environmental Reviews* 19, 333-349. [5] WMO (2012). *Standardized Precipitation Index User Guide* (WMO-No. 1090). [6] UNEP (1992). *World Atlas of Desertification*. United Nations Environment Programme. [7] Middleton, N. & Thomas, D. (1997). *World Atlas of Desertification*, 2nd ed. UNEP. [8] Nicholson, S. E. (2017). Climate and climatic variability of rainfall over eastern Africa. *Reviews of Geophysics* 55, 590-635. [9] Lyon, B. & DeWitt, D. G. (2012). A recent and abrupt decline in the East African long rains. *Geophysical Research Letters* 39, L02702. [10] Thornthwaite, C. W. (1948). An approach toward a rational classification of climate. *Geographical Review* 38, 55-94. [11] Vicente-Serrano, S. M., Begueria, S. & Lopez-Moreno, J. I. (2010). A multiscalar drought index sensitive to global warming: the SPEI. *Journal of Climate* 23, 1696-1718. [12] McKee, T. B., Doesken, N. J. & Kleist, J. (1993). The relationship of drought frequency and duration to time scales. *Proc. 8th Conf. on Applied Climatology*, 179-184. [13] Pagani, M. et al. (2018). Global Earthquake Model (GEM) seismic hazard map; Giardini, D. et al. (1999). The GSHAP global seismic hazard map. *Annali di Geofisica* 42, 1225-1230. [14] Dottori, F. et al. (2016). Development and evaluation of a framework for global flood hazard mapping. *Advances in Water Resources* 94, 87-102 (JRC GloFAS). [15] Giglio, L. et al. (2018). The Collection 6 MODIS burned area mapping algorithm and product (MCD64A1). *Remote Sensing of Environment* 217, 72-85. [16] Raleigh, C. et al. (2010). Introducing ACLED: an Armed Conflict Location and Event Dataset. *Journal of Peace Research* 47, 651-660. [17] Albrecht, R. I. et al. (2016). Where are the lightning hotspots on Earth? *Bulletin of the American Meteorological Society* 97, 2051-2068 (LIS/OTD). [18] Knapp, K. R. et al. (2010). The International Best Track Archive for Climate Stewardship (IBTrACS). *Bulletin of the American Meteorological Society* 91, 363-376. [19] Mora, C. et al. (2017). Global risk of deadly heat. *Nature Climate Change* 7, 501-

506. [20] Guha-Sapir, D., Below, R. & Hoyois, P. EM-DAT: The CRED/OFDA International Disaster Database; UNDRR DesInventar Sendai disaster-loss database.